EXP 13: Implement the Car Price Prediction Model using Python

CODING:

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# Car Price Prediction using Linear Regression

# Training Dataset
X = [2018, 2019, 2020, 2021, 2022]
Y = [500000, 600000, 700000, 800000, 900000]

# Calculate Mean
n = len(X)

mean_x = sum(X) / n
mean_y = sum(Y) / n

# Calculate Slope
numerator = 0
denominator = 0

for i in range(n):
    numerator += (X[i] - mean_x) * (Y[i] - mean_y)
    denominator += (X[i] - mean_x) ** 2

slope = numerator / denominator

# Calculate Intercept
intercept = mean_y - slope * mean_x
```

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# Display Training Data
print("Car Price Prediction Model")
print("\nTraining Dataset:")

for i in range(n):
    print("Year:", X[i], "Price:", Y[i])

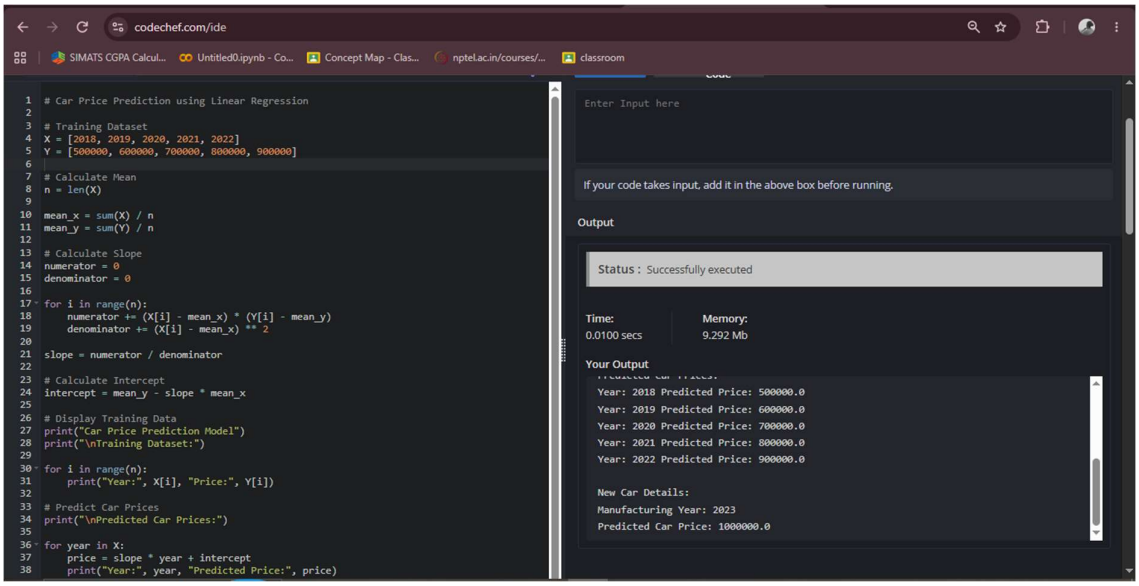
# Predict Car Prices
print("\nPredicted Car Prices:")

for year in X:
    price = slope * year + intercept
    print("Year:", year, "Predicted Price:", price)

# Predict New Car Price
new_year = 2023
predicted_price = slope * new_year + intercept

print("\nNew Car Details:")
print("Manufacturing Year:", new_year)
print("Predicted Car Price:", predicted_price)
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OUTPUT:


codechef.com/ide
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classroom
Car Price Prediction using Linear Regression
Training Dataset
X = [2018, 2019, 2020, 2021, 2022]
Y = [500000, 600000, 700000, 800000, 900000]
Calculate Mean
n = len(X)
mean_x = sum(X) / n
mean_y = sum(Y) / n
Calculate Slope
numerator = 0
denominator = 0
for i in range(n):
numerator += (X[i] - mean_x) * (Y[i] - mean_y)
denominator += (X[i] - mean_x) ** 2
slope = numerator / denominator
Calculate Intercept
intercept = mean_y - slope * mean_x
Display Training Data
print("Car Price Prediction Model")
print("\nTraining Dataset:")
for i in range(n):
print("Year:", X[i], "Price:", Y[i])
Predict Car Prices
print("\nPredicted Car Prices:")
for year in X:
price = slope * year + intercept
print("Year:", year, "Predicted Price:", price)
Enter Input here
If your code takes input, add it in the above box before running.
Output
Status : Successfully executed
Time:
0.0100 secs
Memory:
9.292 Mb
Your Output
Year: 2018 Predicted Price: 500000.0
Year: 2019 Predicted Price: 600000.0
Year: 2020 Predicted Price: 700000.0
Year: 2021 Predicted Price: 800000.0
Year: 2022 Predicted Price: 900000.0
New Car Details:
Manufacturing Year: 2023
Predicted Car Price: 1000000.0